

Mmap Implementation Reflection
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1. Starting with the starter code from Lecture 5, we began by using the [AM335x Technical Reference Manual](#) to find the constant values such as GPIO_BASE_PHYS (Section 2.1 in ARM Cortex-A8 Memory Map), GPIO_OE_OFFSET, and others found in General-Purpose Input/Output (Section 35).
2. After filling in the constants, editing the provided functions to fit our project setup:
 - a. Utilizing the `gpio_map_t` struct, an array of 4 was made to store each of their respective fds, map base, and gpio base values.
 - b. As such, it was no longer needed to pass around a `gpio_map_t* m` since the object was now being accessed through the `gpio_control.c` file directly.
 - i. Furthermore, the `gpio_map_close` function was also edited to access the array directly.
3. To then complete the set function, we are given two constants `GPIO_SETDATAOUT_OFF` and `GPIO_CLEARDATAOUT_OFF`, which we can utilize to either set the register bit to 0 or 1, depending on whether the light should be on or not. Then, utilizing the `reg32` provided function, we can get the 32-bit registers and set the corresponding bit to whichever we may need (utilizing bitwise operations as done in the [docs](#) as well).
4. Following this is the set direction function, where we say whether the pin is going to be an in or out direction pin. Utilizing the constant `GPIO_OE_OFFSET`, we can set the value to 0h as stated by the manual to set the pin to an output (for the LED).